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Research Article

Improving Road Safety in Smart Cities using Machine Learning Techniques

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Abstract: Road safety is a serious concern in urban environments and smart cities, involving the well-being of all road users, with a special emphasis on those considered vulnerable such as pedestrians, bicyclists, and motorcyclists among others. Being a difficult challenge, it becomes evident that the traditional methods of road safety are not sufficient to address these challenges and needs a more proactive and adaptive solution. This study aims to analyze traffic accident data, identify factors behind severe injuries, and develop predictive solutions using Machine Learning (ML). For this study, data were collected from the Public Safety Data Portal in Toronto and the District Emergency Office in Rawalpindi, Pakistan (RTA). The data were preprocessed by applying different data preprocessing methods using Python. A number of popular ML and Deep Learning(DL) models have been trained and tested on the preprocessed datasets. Results showed that the XGBoost exhibited excellent performance with an accuracy of 75% on KSI dataset without under-sample and hyperparameter tuning methods. Random Forces achieved high accuracy 99% after applying the Grid Search method of hyperparameter methods and undersample technique Moreover, the current study has utilized the Association Rule Mining technique to determine the underlying hidden factors from both dataset that lead to collisions and fatal or major injuries. The extracted rules revealed that certain factors, such as over speeding, aggressive driving, pedestrian collisions, disobeying traffic rules, lost control, driver's inattentiveness and the absence of traffic controls at major roads and intersections are associated with a higher risk of fatal or major injuries. Furthermore, the study provided a comprehensive comparison of factors contributing to severe collisions in Pakistan and Canada. It found that there is a need for targeted enforcement in both countries including stricter licensing and education initiatives for young drivers about responsible driving behaviors. The proposed framework can be deployed in a real environment for improving road safety in urban environments and Smart cites

Keywords: Road Safety; Smart Cities; Machine Learning; Association Rule Mining; Mobility Prediction;

1. Introduction

Road accidents are a significant public health issue worldwide. Each year, 1.9 million people die due to traffic accidents and the majority are young people aged 5 to 29 years old [20]. Road accidents analysis and safety is a hot research area for finding the underlying cause of road accidents and to implement effective preventive measures [21].

Many researchers have discussed road accident issues and the literature study shows that the traditional techniques are not sufficient to capture the dynamic and complex nature of road accidents [21]. The need for a more proactive and responsive solution is crucial. Moreover, the necessity for an enhanced traffic safety system, powered by Artificial Intelligence (AI) and Machine Learning paradigms, is becoming increasingly important. By utilizing the capabilities of Machine Learning (ML), more intelligent and adaptive systems can be developed for safer and efficient transportations. This can enhance mobility in the urban environment without overshadowing the possible risk of accidents [21].

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1.1. Road Accidents

A road traffic accident is an unintentional event occurring on a public road with another vehicle. As a result, injury, death, property damage, or environmental harm occurs [22]. Road accidents are a main cause of kids and young people's death worldwide. The World Health Organization (WHO) report shows approximately 1.3 million people die annually due to road accidents. Moreover, half of these fatalities are road users such as pedestrians, bicyclists, and motorcyclists. In addition, an estimated 20 to 50 million people suffer non-fatal injuries each year, often resulting in permanent impairments [23].

1.2. Significance of Predicting Injury Severity

Enhancing road safety is an important research topic, because it is one of the leading reasons for fatalities in the world. There are a number of factors that lead to severe or fatal injuries and analysis of such factors are important so that proper preventive measures can be taken to reduce the fatalities rate due to road accidents [20].

The literature study shows that machine learning based predictive models have been widely used for the safety of human life especially in the field of road safety and accidents. Moreover, the prediction of injury severity appears as a keystone in developing effective preventative road safety tactics [24]. The ability to predict the probability of an accident gives us a proactive position, enabling early interventions and limiting the aftermath's impact. A well-informed decision can be made by analyzing the real-time data to avoid road accidents [24]. The accurate predictive models for road safety can be used in several areas, such as in Emergency Medical Services (EMS) to prioritize cases, dispatch appropriate resources [25]**.** Moreover, early prediction enables the healthcare providers to timely facilitate patients by allocating beds, ordering tests, and staff availability [25].

1.3. Machine Learning (ML) Techniques in Predicting Injury Severity

The Road Accident Analysis using Machine Learning (ML) technique is an important study that applies advanced ML techniques to analyze road traffic accidents and predict injury severity. ML is widely being used to build prediction models from traffic datasets. Furthermore, these approaches have the ability to find the hidden pattern from the datasets [12].

The road accident data is multidimensional and quite heterogeneous, the traditional statistical analysis may not be able to identify hidden patterns and relationships in the data that are important for predicting the likelihood of fatal or major injuries. However, machine learning (ML) techniques can provide a solution to this problem. In the domain of road safety research, the incorporation of advanced machine learning techniques offers a viable route for understanding and tackling the complexities connected with road traffic accidents [21]. Decision Trees (DTs) is one of the most widely used machine learning algorithms that is used to create trees and find prediction rules based on existing data [37]. Soheil Rezashoar, Ehsan Kashi and Soheila Saeidi (2023) have also utilized the ML technique for the prediction of traffic accident severity. They have provided a comparison of several algorithms in their work and concluded that Naive Bayes, Support Vector Machine and Neural Network have predicted the accident severity with reasonable accuracy [38]. Furthermore, they have also stated that it is not necessary to define the relationship between variables while implementing ML models and these models are more suitable where accuracy matters than the associations. The previous study used traditional ML techniques and they cannot investigate hidden factors of road safety. The current study has applied multiple ML techniques and advanced Deep Learning (DL) methods as analytical learning approaches intended to predict the injury severity level and to find out the hidden associations among the critical factors. The study aims to use predictive and descriptive machine learning approaches to get insights into traffic accident datasets.

1.4. Innovation of the Current Study

This study has made several contributions and key enhancements in the field of transportation and road safety research. . Firstly, this study has performed cross country analysis of the injury severity level of the victims by comparing accident datasets from the District Emergency Office in Rawalpindi, Pakistan (RTA) and Toronto Canada. This offers both context specific and universal insights which are rarely investigated in the literature. Moreover, the current study has implemented various popular ML models including Random Forest (RF), Support Vector Machine (SVM), XGBoost, Bidirectional Long Short-Term Memory (BiLSTM), Attention-Based Long Short-Term Memory (ATTR-LSTM), K-Nearest Neighbors Long Short-Term Memory (KNN-LSTM), Wavelet Convolutional Neural Network Long Short-Term Memory (W-CNN-LSTM) on the road traffic accident datasets. It is concluded that the XGBoost is the best performing

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model for accident injury severity prediction having accuracy of 75% on KSI dataset. and RF model achieved 99% accuracy during Hyperparameter tuning and undersampling process.

Secondly, this research has integrated the best performing model with Association Rule Mining that not only improves the predictive accuracy but also uncover certain hidden factors that are associated with a higher risk of fatal or major injuries. These factors include over-speeding, aggressive driving, pedestrian collisions, disobeying traffic rules, loss of control, driver's inattentiveness, and the absence of traffic controls at major roads and intersections. Furthermore, the study has provided a comprehensive context specific comparison of these associations and identified that Pakistan needs to ensure the proper measures to enforce the use of helmets for bike riders along with improving the lighting conditions. On the other hand, Canada should adopt the proper measures to ensure pedestrian and cyclist safety. Both of these countries need to focus on strict licensing and education of young drivers. This study considered the following research questions.

Question 1: Which machine learning or deep learning model provides the best trade-off between accuracy and computational efficiency for injury classification?

Question 2: What are the underlying hidden factors that contribute to fatal or major injuries?

The rest of this paper is organized as follows. **Section 2** discusses the related work**.** The research methodology is discussed in **Section 3. Section 4** describes the experiment and discussion. **Section 5** presents the results and discussion.

2. Related Work

Considerable research has been conducted in the past to investigate road traffic accidents in different directions. Past studies have used different Machine Learning (ML) techniques on different datasets with varying numbers of input features to investigate the relationships between these features. Previous research has been conducted with the aim of building a prediction model that is capable of automatically classifying the accident severity level.

Ahammad et al. have developed models that are based on Random Forest (RF), Naïve Bayes, Gradient Boosting Classifier (GBC), Logistic Regression (LR), and K-Nearest Neighbor (KNN). The research has used road traffic accident data from the UK. The authors have suggested that the Random Forest classifier has achieved the best results with an accuracy of 63%, followed by the GBC with an accuracy of 62%. Gradient Boosting has taken a longer time to build the model as compared to Random Forest, but it has achieved the highest F1 score and AUC among all others Moreover, the research has suggested that the severity of an accident is related to many factors, not any single factor contributes to it. It is highly dependent on temporal data including weather, light conditions, road surface, traffic signal, speeding among others [14].

Kuyumcu et al. Conduct a study to find out the relationship between factors that increase the chance of accidents using machine learning methods. This research has applied the association rule mining techniques on traffic accident data. Their dataset contains 6 years of collision data of the city Sakarya, Turkey from 2015 to 2020 having 23, 551 entries and 26 attributes. This research has found a lot of association rules, but they had taken the lift as a measure to consider the most important rules. It has been concluded that speeding is highly correlated with the injury severity, rollover and run off road leads to major vehicle damage. There exist seasonal patterns in the dataset such that fatal injuries are most common in spring and summer seasons due to high speed of vehicles. Drivers with low educational level tend to be involved more in collisions due to rule violations and side collisions. It has also been found out that the young drivers lead to severe accidents due to over speeding whereas the middle-aged drivers are mostly involved in the side collisions [15]

Neil and Datu [16] have uncovered the insights of the traffic accidents happening in the Angeles city of Philippines. They applied the association rule mining technique on 1438 records of traffic collisions from January 2015 to June 2018. The data was collected from Angeles City Traffic Management and Enforcement Unit (ACTMEU). The research conducted to find out the most common characteristics of the suspected motorists and their attributes that contribute mostly to road accidents. Confidence, support and lift are the metrics that have been used to evaluate the association rules in the study. They have revealed from association rules and descriptive analysis that the majority of the road accidents involve male drivers with age range between 15 to 29. Balibago city is the most prone to collisions from 8 PM to 12 AM and on Sunday. However, the association rules point out that road accidents are more likely to happen from

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8 Pm to 12 AM and on Saturday. A further analysis shows that Balibago having the greatest number of accidents does not have the higher probability of having collisions during the 8Pm to 12Am on Saturday [16].

Emu M et al. developed a model that predicts the consequence of a collision as fatal or non-fatal. The focus of their study was the prediction of collision severity with the help of machine learning algorithms. For developing the required model, this research has used four machine learning methodologies that include Support Vector Machine, Random Forests, K-Nearest Neighbor, and Convolutional Neural Network. The study has used accuracy, recall, precision and F1-score as performance measures for each method. The accuracy of all of these methods ranges from 69.38% to 75.56%. The KNN has the lowest accuracy of 69.38% and the CNN performed significantly higher than all of these methods with an accuracy of 75.56%. CNN also has better precision (75.74%) and recall (75.45%) than the other methods. So, these results show that Convolutional Neural Network (CNN) achieves the best performance in the prediction of accident seriousness by using both soft and majority voting [6].

Yadav et al. identified the factors that lead to road accidents. The authors have used the ML techniques to investigate the factors that impact the road accidents. For this purpose, they have used the Killed or Seriously injured (KSI) dataset provided by the City of Toronto Police in their Open Data Portal. This dataset contains all the information of traffic accidents that have been reported from 2007 to 2017. The research has applied both supervised machine learning techniques including Decision Trees, K-Nearest Neighbor (KNN), Naïve Bayes and AdaBoost and unsupervised machine learning techniques including K-mean clustering algorithm, CART and ROC value. This research has achieved 81.5% performance with the CART algorithm and AdaBoost also achieves the best performance among these. It has been concluded that most of the accidents that lead to fatal injuries happen in the east Toronto region and the victims are mostly pedestrians. Aggressive driving (62.9%) is the major cause of accidents, then over speeding (21.4%), red light signal violation (10.4%) and then alcohol (5.3%) [1].

Shanshal D et al. have developed a prediction model that identifies the patterns in severe and fatal collisions happening in Toronto. This research tried to predict the fatalities of individuals including only drivers, pedestrians, and cyclists using data analytics and machine learning techniques. The KSI dataset provided by the Toronto Police Services is used in this study. This research has used the data mining approach of association rules, classification algorithms including Lasso Regression and Random Forest. The experimental results reveal that the Random Forest technique has outperformed with accuracy of 80% for drivers and pedestrians and 89% for cyclists. . The research has used the apriori algorithm for finding association rules and it is concluded that the factors of aggressive driving, failing to yield right of way, improper turns, and driver's inattentiveness lead to severe collisions. Several environmental conditions such as clear and dry weather in summer and spring seasons lead to more accidents. The cyclist in the months of June and July gets fatal injuries. In the case of rainy weather, wet surfaces, and dark light causes pedestrians's risk of fatal or major injury. The cyclist gets fatal injuries when the driver sideswipes cyclists while driving in the same direction, a motorist turning left across the cyclist's path and the cyclist struck by the opened vehicle door [2].

Gan J et al. [7] used the Deep Forest (DF)algorithm based on the decision tree ensemble for the prediction of severity of traffic accidents... This research used the UK road safety dataset in 2016 obtained from the Kaggle website. For the validation of their proposed model, this research has incorporated other famous machine learning algorithms including DNN, Random Forest, LightGBM, XGboost, K-Nearest Neighbor (KNN) and decision trees. The results show that Deep Forest has outperformed their counterparts with accuracy of 90.69%, the recall is 0.92, and the highest F1 score and ROC of 0.91 and 0.93 respectively. The high prediction accuracy of Deep Forests algorithms reveals that it can be used more effectively for the prediction of accident severity [7].

Hébert et al. [5] developed a machine learning model to predict the likelihood of collisions at various locations in the city of Montreal. In this study, authors have used three machine learning models including Random Forest (RF), Balanced Random Forest (BRF), and the XGBoost algorithm. The research reveals that the RF, BRF, XGB, and baseline model have gained 0.918, 0.916, 0.909, and 0.874 area under the ROC curve. This shows that RF and BRF have almost the same performance and perform better than the XGBoost and baseline model. The results show that RF has the best performance among all of these models, and it has successfully detected 85% of road collisions with a false positive rate of 13%. This model could be used to identify the most dangerous road segments every hour, so the necessary actions could be taken in order to avoid the risk of accidents [5].

Labib et al. [8] conducted a study to analyze and predict the severity of road accidents in Bangladesh using machine learning techniques. This research has used four supervised machine learning algorithms i.e., Decision Trees, K-Nearest Neighbors (KNN), naive Bayes, and Adaptive Boosting (AdaBoost). The reason for using these models is due to their proven accuracy. The intensity of accidents has been classified into four categories including fatal, grievous,

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simple injury, and motor collision. They have used a traffic accident dataset of Bangladesh from 2001 to 2015 that contains 43000 records. They have performed two experiments on the dataset and calculated the accuracy as 71%, 67%, 80%, 80%, precision as 68%, 68%, 63%, 68%, and F1 score as 71%, 67%, 71%, and 73% for DT, KNN, Naive Bayes and AdaBoost respectively. In the second experiment, the dataset was divided into two classes called Fatal and Grievous, and found that the accuracy of DT and KNN slightly increased. The performance of AdaBoost increases the F1 score and precision increase from 73% to 74% and 68% to 75% respectively. It is concluded from the above experiments that AdaBoost achieves the best performance among all of these machine learning techniques [8].

Parathasarathy et al. [9] develop several predictor models to predict the accident type. They have used machine learning approaches for their study that include K-Nearest Neighbor and Support Vector Machine algorithms, have been developed to predict the accident types [9]. The model proposed by them uses a combination of both KNN and SVM algorithms that have better accuracy as compared to the separate algorithms. Initially, the KNN algorithm was used to classify the dataset and predict the type of accident, which was then along with other factors used to train the SVM algorithm. They obtained a dataset from the UCI Repository that contains 1000 records with 12 variables. It is found that the KNN and SVM both give an accuracy of 60% whereas the hybrid approach used in this study gives an accuracy of 92%. So, it is concluded that the use of this hybrid approach proved more accurate in finding the accident class as compared to the individual use of the algorithm [9].

Kumeda et al. [10] conducted research to find out the major and key factors that cause traffic accidents, so that these factors can be used to form policies to reduce the traffic accident rate. The authors have used a dataset from UK road traffic accidents of the year 2016 that contains 12 attributes and 555 records. They have applied 6 machine learning algorithms that include Fuzzy-FARCHD, Random Forest, Hierarchical LVQ, BRF Network, Multilayer Perceptron and Naïve Bayes on this dataset and obtained an accuracy of 85.94%, 83.42%, 82.16%, 84.14%, 79.27%, and 80.90% respectively. It can be seen that the fuzzy-FARCHD classifier has obtained highest accuracy. The analysis shows that most of the causality's ages lie between 20 to 40 years, most of them were male and the involved vehicle type is a car and mostly happens in daylight [10].

AlMamlook et al. established different models for classifying the injury severity during road accidents and to identify the factors behind traffic accident severity. This research has obtained the dataset from the Office of Highway Safety Planning (OHSP) that contains 271,563 road crashes that occurred in Michigan during the period 2010 to 2016. They have built several models using different machine learning algorithms and compared their performances to select the most suitable model. They have applied AdaBoost, Logistic Regression, Naïve Bayes (NB), and Random Forests (RF) algorithms on the data set. It has been concluded that the RF algorithm has achieved the best performance with an accuracy of 75.5% than LR with 74.5%, NB with 73.1%, and AdaBoost with 74.5% accuracy. This is because Random Forest has a good ability to resist noise as compared to other machine learning algorithms [11].

Gaurav and Zunaid Alam [4] conducted a study with the aim of reducing the road accidents in India. This research to reduce the road accidents in India [4] data mining approach of association rule on the dataset provided by National Highways Authority of India (NHAI) has been used and analyzed for finding the important patterns of accidents. The dataset contains the data from Panipat to Jalandhar. The application of association rule mining algorithm reveals that there is a high percentage of accidents in Punjab. Most of the accidents are injury based involving petrol vehicles as compared to other vehicles. The accident rate is high on four lane highways as compared to two lane highways. Late night and morning sessions are more susceptible to collisions. Over speeding, rear end collisions and collisions due to the fault of other drivers while overturning has strong confidence. Most of these accidents are only injury based and a very few serious accidents occur. So, it is suggested to avoid over speeding to further decrease the number of accidents [4].

Watkins E et al. [3] have analyzed the traffic scenarios that could lead to fatal or major injuries or even deaths in their study. The main aim of their study was to identify the occurring patterns of fatalities and injuries to recommend some preventive measures that could help in reducing these occurrences [3].This research has used The National Collision Database (NCDB) provided by Transport Canada in this study for analysis. This research has used the data mining techniques of decision trees and neural networks in order to fulfill the research purposes by using Microsoft tools. The research has done various tests after data preprocessing and finds out that: If the driver's age is above 60 then there are six times more chances of incidents in which a person conceived death as compared to young drivers. A curved tunnel or underpass has 57.38% chances of single vehicle incidents. There is an 81.60% chance of at least one fatality in collisions

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that happen at railway crossings where signs are only being used to control the traffic. It has also been found out that the new drivers feel nervousness when driving on a circular path or on express freeways that results in more collisions. The vehicle manufacturing year has an effect on the fatality rate, there is a general decrease of fatalities with the latest models.

Most existing research uses traditional models to predict road traffic accidents and few combine both predictions and find hidden factors but these models achieved moderate accuracy, because road accidents are a major cause of economic and human loss in all countries, therefore, it is a great challenge for health and transportation departments. Current study used traditional and advanced deep learning models to predict and hidden factor of road accident with high accuracy,

3. Methodology

In the current study, we applied conventional and advanced Machine Learning (ML) techniques for analyzing road traffic accident dataset**.** This study consists of two phases: the first phase, predicting the injury severity in the collision and the second phase consisting of finding the factors that lead to severe collisions using ML algorithms**.** The proposed prediction framework has been divided into five steps i.e., data collection, data preprocessing, finding the association rules, data splitting and model training, and lastly model evaluation. **Fig. 1** shows steps that have been followed in the proposed framework.

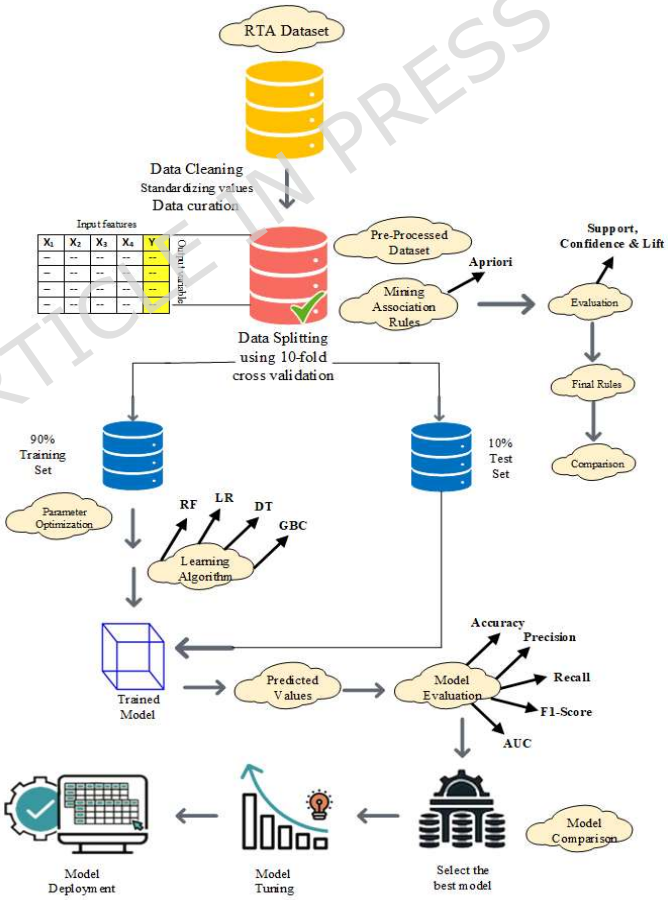

Fig 1 Proposed Framework

3.1. Data Description

The current study used two datasets, the first one is the Killed or Seriously Injured (KSI) dataset obtained from the Public Safety Data Portal provided by Toronto Police Services [13]. This dataset contains all of the traffic collisions that happened from 2006 to 2022. This dataset includes 57 variables and 18,194 rows. After performing, preprocessing steps

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16,578 rows and 45 features left. These variables can be classified as individual attributes and collision-related attributes, the description of these variables is shown in **Table 1**.

Table 1 Input features of the KSI dataset

Name	Description	Variable Type
YEAR	Year in which collision occurs	Numerical
MONTH	Month in which collision occurs	Numerical
DATE	Date of collision occurrence	Numerical
DAY	Day of collision occurrence	Nominal
TIME	Time of collision occurrence	Numerical
ROAD CLASS	Road Classification	Categorical
DISTRICT	City District	Categorical
LOCCOORD	Location Coordinates	Categorical
ACCLOC	Collision Location	Categorical
TRAFFCTL	Traffic control type	Categorical
VISIBILITY	Environment condition	Nominal
LIGHT	Light condition	Nominal
RDSFCOND	Road surface condition	Nominal
ACCLASS	Class of accident	Ordinal
IMPACTYPE	Initial impact type	Categorical
INVTYPE	Role of the person involved in the collision	Categorical
INVAGE	The age range of the involved person	Nominal
INJURY TYPE	Level of injury the involvement sustains	Ordinal (Target Variable)
VEHTYPE	Type of Vehicle	Categorical
MANOEUVER	Vehicle Maneuver	Categorical
DRIVACT	Apparent Driver Action	Categorical
DRIVCOND	Driver Condition	Categorical
PEDTYPE	Pedestrian Crash Type – detail	Categorical
PEDACT	Pedestrian Action	Categorical
PEDCOND	Condition of Pedestrian	Categorical
CYCLISTYPE	Cyclist Crash Type – detail	Categorical
CYCACT	Cyclist Action	Categorical
CYCCOND	Cyclist Condition	Categorical
PEDESTRIAN	Pedestrian involvement in collision	Binary (Yes/No)
CYCLIST	Cyclist involvement in collision	Binary (Yes/No)
AUTOMOBILE	Automobile involvement in collision	Binary (Yes/No)
MOTORCYCLE	Motorcycle involvement in collision	Binary (Yes/No)
TRUCK	Truck involvement in collision	Binary (Yes/No)
TRSN_CITY_VEH	Transit or City Vehicle Involved in Collision	Binary (Yes/No)
EMERG_VEH	Emergency Vehicle Involved in Collision	Binary (Yes/No)
PASSENGER	Passenger Involved in Collision	Binary (Yes/No)
SPEEDING	Speeding Related Collision	Binary (Yes/No)
AG_DRIV	Aggressive and Distracted Driving Collision	Binary (Yes/No)
REDLIGHT	Red Light Related Collision	Binary (Yes/No)
ALCOHOL	Alcohol Related Collision	Binary (Yes/No)
DISABILITY	Medical or Physical Disability Related Collision	Binary (Yes/No)

The second dataset is collected from the District Emergency Office (Rescue 1122) Rawalpindi **[17]**. This dataset contains all the traffic accident events that happened in the District Rawalpindi from January 2020 to July 2023. The dataset includes 25 variables and 46,189 records, after performing preprocessing and feature extraction steps 46,187 rows and 27 features left. The description of features is described in **Table 2**.

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Table 2 Input features of the RTA dataset

Name	Description	Variable Type
Year	Year in which collision occurs	Numerical
Month	Year in which collision occurs	Numerical
Date	Year in which collision occurs	Numerical
Day	Day of collision	Nominal
Time	Time of collision	Nominal
Nature of Weekday	Specifies working day or weekend in which collision occurs	Nominal
Season	Season in which collision occurs	Nominal
Light	Lighting condition during a collision	Nominal
Weather	Weather conditions	Nominal
Total Patients	No of patients involved in a collision	Numerical
Gender	Victim's gender	Nominal
Age Range	Victim's age	Nominal
Education	Victims' education level	Nominal
Acclass	Class of accident	Ordinal
Injury Type	Level of injury severity	Ordinal (Target Variable)
Cause	Reason of accident	Nominal
Patient Status	Condition of victim	Ordinal
Bicycle Involved	Bicycle involvement in collision	Binary (Yes/No)
Bikes Involved	Bikes involvement in collision	Binary (Yes/No)
Buses Involved	Buses involvement in collision	Binary (Yes/No)
Cars Involved	Car's involvement in collision	Binary (Yes/No)
Cart Involved	Cart involvement in collision	Binary (Yes/No)
Rickshaws Involved	Rickshaw's involvement in collision	Binary (Yes/No)
Tractor Involved	Tractor involvement in collision	Binary (Yes/No)
Trains Involved	Trains involvement in collision	Binary (Yes/No)
Trucks Involved	Trucks involvement in collision	Binary (Yes/No)
Vans Involved	Vans involvement in collision	Binary (Yes/No)

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3.2. Proposed Framework

The current study has proposed an enhanced framework for analyzing road traffic accidents using ML and DL techniques. It consists of two phases: 1) In the first phase, injury severity is predicted with traditional and advanced DL models2) In the second phase, hidden factors related to injury severity are identified. The proposed methodology for this study has been divided into five steps as follows:

The first step is data collection, in this step, we have obtained two datasets, the first one is the KSI dataset obtained from Toronto Police Services web portal and the second one is the RTA dataset obtained from the Office of Punjab Emergency Services, Rescue 1122 Rawalpindi [17,13]. The second step is data preprocessing. In these steps, the study has used several techniques including data cleansing, data transformation, data imbalance and normalization to prepare data for the ML model. For data transformation we used label encoding and for normalization we used standard scaler methods.

The third step is finding the association rules, the study has used the apriori algorithm in order to find out the hidden patterns and association among several factors that leads to severe collisions along with the comprehensive comparison.

The current study uses Random Forest, SVM, XGBoost, BiLSTM, ATTR-LSTM, KNN-LSTM, W-CNN-LSTM K-Nearest Neighbors Long Short-Term Memory for Predicting injury severity in road accidents. These Machine Learning classifiers are widely used for supervised classification tasks.
The fifth step consists of model evaluation, in this step the developed models were compared and evaluated against some performance measures including accuracy, recall, precision, f1 score and AUC [31].

3.3 Code Availability

The current study machine learning models are trained and evaluated in the latest version of Python and implemented in the Google Colab environment. The Python code of this publication is publicly available at the GitHub repository. So that reproducibility and transparency of code is maintained. The Digital Object Identifier (DOI) has been archived at [39]. Moreover, there is no restriction to access the code.

4. Experiment and Discussion

In this section, collision injury severity level is investigated using various features related to accidents. Both predictive and descriptive data mining tasks were performed to address current study research questions. The analysis involved the implementation of several MLand DL Techniques and association rule mining techniques.

4.1. Data Visualization

We performed data visualization using Python to further understand the experimental datasets. These visualization results show the importance of input attributes in predicting the injury severity.

Firstly, the study has analyzed the KSI dataset, the target variable has been analyzed against different input features. **Fig. 2** shows that the overall trend of accidents has been reduced over the years.

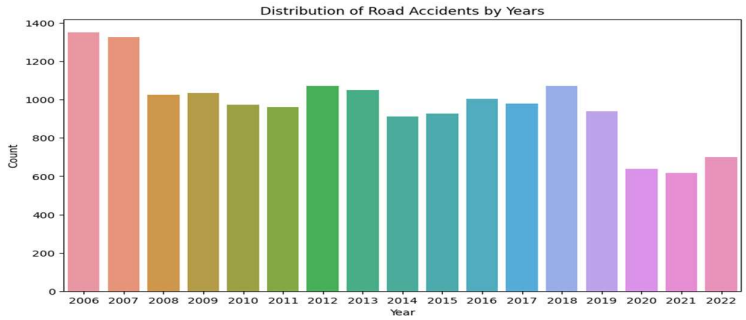

Fig 2 Number of Collisions Happened over 17 Years (2006-2022) in KSI dataset

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Upon delving deeper into the dataset, a distinct pattern emerges, highlighting Fridays as the day with the highest frequency of collisions, particularly those resulting in Fatal or Major injuries as shown in **Fig. 3**. In contrast, Sundays exhibit the lowest ratio in this regard.

Fig. 3 represents this observation and underscores the significance of temporal factors in influencing the severity of traffic accidents, with the end of the work week seemingly contributing to a heightened risk scenario.

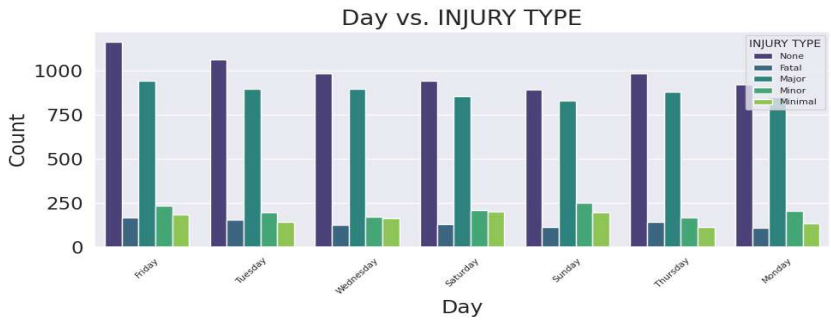

Fig 3 Frequency of Collisions and the Injury type against Weekday on KSI dataset

Next, the analysis of the Road Traffic Accident (RTA) dataset spanning from 2020 to July 2023, **Fig. 4** shows a clear pattern that becomes apparent when looking at the yearly frequency of accidents. Between 2020 and 2022, there is a clear and worrisome increase in the frequency of accidents, suggesting a significant rise in road occurrences throughout this time frame.

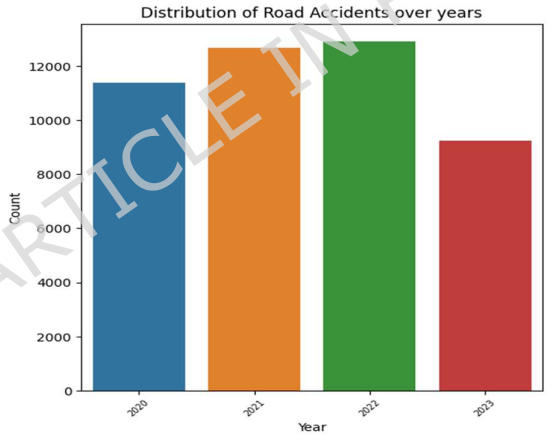

Fig 4 Frequency of Collisions in RTA Over Years (2020-July 2023)

The data indicates that each subsequent year had a greater number of collisions than the previous year, implying a possible worsening of road safety issues throughout this period. Since the information only covers up until July 2023, the yearly results for that year are still incomplete and may not provide an accurate representation of the general pattern.

When examining the gender-specific aspects of injury severity in road traffic accidents, an obvious pattern can be seen from the graph shown in **Fig. 5**. The significant predominance of males in both the occurrence of accidents and the severity of injuries incurred is a convincing depiction of gender-related trends in road safety occurrences.

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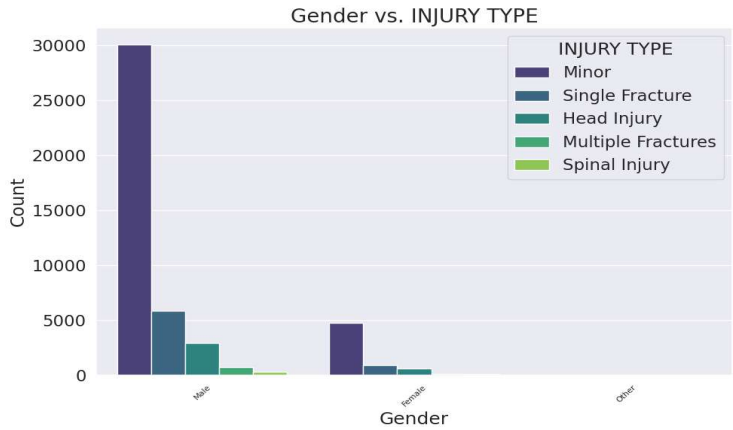

Fig 5 Frequency of Collisions in RTA against Gender

The significant majority of males who suffer from small injuries and single fractures highlight the necessity for specific procedures and focused awareness campaigns. This finding stimulates a thorough investigation into the possible elements that are influencing these trends.

Fig. 6 represents an analysis of the correlation between age and road traffic accidents; it indicates a significant concentration of collisions and injuries among those having an age range between 20-30. This particular age group is found to be the most vulnerable to road accidents, indicating the importance of implementing specific safety measures and educational programs for this population.

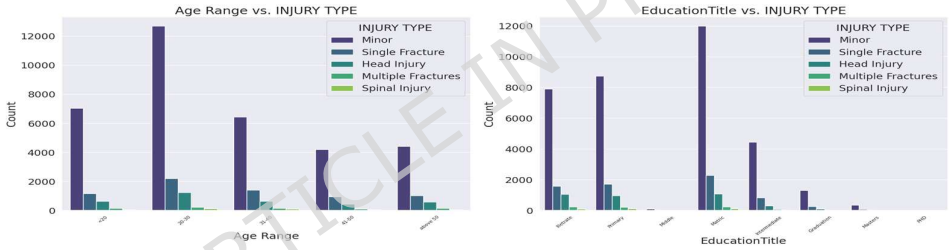

Fig 6 Frequency of Collisions in RTA against Age Range and Education Title

Shifting emphasis to the association between education levels and accident engagement, a remarkable tendency arises. Contrary to expectations, **Fig 6** shows that persons with educational levels equal to matric have the largest representation in traffic crashes, followed by those with a primary school education and then, shockingly, illiterate individuals. This surprising discovery contradicts conventional assumptions and motivates a closer examination of the factors influencing accident rates among various educational levels.

4.2. Statistical Analysis

This study has also incorporated statistical analysis before applying the ML techniques on both datasets to identify how much the input features are correlated with our targeted attribute. The study has conducted the Chi-Square Contingency test along with the calculation of Cramer's V. The Chi-Square and Cramer's phi are the most suitable statistical techniques according to the nature of these datasets [19] . After conducting the analysis each independent variable got Chi-Square statistic, p-value, degree of freedom, and Cramer's phi. These values reflect the strength and significance of the correlation between the tested pair of attributes. The topmost important correlations of the KSI dataset are shown in **Table 3**.

Table 3 Statistical Analysis on KSI Dataset

Variable	Chi-Square Statistic	P-value	Degree of Freedom	Cramer's V
INVTYPE	8754.868189	0.000	64	0.363452
ACCLASS	6567.699552	0.000	8	0.445067

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VEHTYPE	4435.858836	0.000	120	0.287356
PEDESTRIAN	954.76729	0.000	4	0.239984
TRAFFCTL	108.55384	$3.34E^{-09}$	36	0.040502
DRIVCOND	1096.557247	0.000	36	0.172358
RDSFCOND	90.977844	1.46E-07	32	0.037067
ROAD_CLASS	101.925096	3.23E-08	36	0.03959
LIGHT	75.68876	2.12E-05	32	0.033785
SPEEDING	261.805108	1.86E-55	4	0.125668

The statistical analysis on KSI dataset shows that INVTYPE, ACCLASS, and PEDESTRIAN are highly significant and have strong association with injury severity type. The VEHTYPE, DRIVCOND and SPEEDING are also highly significant but have a moderate association, the TRAFFCTL, RDSFCOND, ROAD_CLASS and LIGHT have a high significance and weak association with the targeted attribute.

Similarly, the same statistical analysis conducted on the RTA dataset, **Table 4** shows the topmost significant correlations.

Table 4 Statistical Analysis on RTA Dataset

Variable	Chi-Square Statistic	P-value	Degree of Freedom	Cramer's V
Patient Status	14305.29379	0.000	8	0.393526
ACCLASS	5258.443386	0.000	4	0.337418
Total Patients in Emergency	509.534073	0.000	44	0.052517
Trains involved	557.56488	0.000	4	0.109872
Trucks involved	701.024602	0.000	12	0.071129
Age Range	183.039109	0.000	16	0.031476
Cause	283.857996	0.000	24	0.039198
Education title	201.455444	0.000	28	0.033022
Gender	111.796691	0.000	8	0.034789
Light	105.60159	0.000	4	0.047816

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The statistics on RTA dataset shows that Patient Status and ACC LASS variables exhibit a very strong association and will provide a high contribution in prediction. The variables of total patients in emergency, trains involved and trucks involved show a moderate association and are statistically significant factors in predicting the injury severity. Lastly, age range, cause, education title, gender and light variables provide weaker association and have statistically small impact that suggests that alone they may not be a strong predictor of injury severity.

4.3. Experimental Setup

The experimental setup used for this study includes Machine Learning (ML) and Deep Learning (DL) models using Python. Python is a popular ML programming language and provides a large set of libraries which can efficiently handle large datasets. These libraries include Numpy, Pandas, Matplotlib, SKLearn, Math, Tensorflow decision forests, and Apyori. The datasets used include the KSI dataset of Public Safety Data Portal from Toronto police services and the RTA dataset from Punjab Emergency Services (**Rescue1122, Rawalpindi**) [17]. These datasets comprise of 45 and 29 attributes and 16578 and 46187 records respectively. The proposed technique has used 57 input features with target variable "Injury Type" for model training. To enhance model scalability, a 2-fold cross validation technique has been

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applied. The performance evaluation criteria of this study include accuracy, precision, True Positive Rate, Area Under the Receiver Operating Characteristic Curve (AUC-ROC), and F1-score. These criteria are computed as:

TPR = TP/TP+FN
Precision = TP/TP+FP
Accuracy = TP/TP+TN+FP FN
F1 score = 2PR/P+R

True Positive Rate (TPR), also known as **recall**, measures the proportion of actual positive cases correctly identified (**TP / (TP + FN)**) (Sokolova & Lapalme, 2009). **Precision** quantifies the accuracy of positive predictions (**TP / (TP + FP)**) (**Accuracy** represents overall model correctness (**(TP + TN) / (TP + TN + FP + FN)**) (. The **F1-score**, a harmonic means of precision and recall, balances false positives and false negatives (**2 × (Precision × Recall) / (Precision + Recall)**).

In ML, the TPR (True Positive Rate) is also known as recall or sensitivity and indicates the percentage of actual positive values that are correctly predicted by the model. Finally, the ROC curve plots the TPR of the model [11]. True Positive (TP) shows Correct predictions of true events. True Negative (TN) indicates Correct predictions of false events or the model correctly predicted a negative class. False Positive (FP) means incorrect predictions of true events or the model incorrectly predicted a positive class. False Negatives (FN) show Incorrect predictions of false events or the model incorrectly predicted a negative class [35]. The ROC curve is used to graphically represent the performance of binary classification models which plot the TPR and FPR at different threshold settings [36]. Other experimental settings shown in **Table-5**.

Table 5- Experimental Setup

Aspect	Detail
Data splitting	2 fold validation ,n_splits=2, shuffle=True, random_state=SEED, SEED=42
Environment	Kaggle notebook, Google Cloud GPU VM
Tools/Frameworks	Python 3.10.18, NumPy 2.2.6, Pandas 2.3.0, Scikit-learn 1.5.2, XGBoost 2.1.1, TensorFlow 2.18.0, Imbalance-learn 0.12.3
Dataset	KSI dataset[Toronto Police Services web portal], RTA dataset [Office of Punjab Emergency Services, Rescue 1122 Rawalpindi Pakistan]
Dataset Size	16,578 records, 42 features
Class Distribution	[6953, 931, 6149, 1422, 1123] (imbalanced)
Preprocessing	Data cleaning, missing value handling, normalization, feature encoding
Sampling Techniques	SMOTE (Synthetic Minority Over-sampling Technique), Random Undersampling, and K-Fold (K=2) cross-validation
Parameter Tuning method	Grid Search
Hyperparameters Tuned	XGB: max_depth: 3, n_estimators: 50 RF: max_depth: None, n_estimators: 50 SVM: C: 1, kernel: rbf
Evaluation Metrics	Accuracy, F1-macro, F1-micro

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Validation techniques	Stratified train-test split combined with K-Fold (K=2) cross-validation

This division of datasets has been merged into 6 groups as shown in **Fig. 7**, to acquire association rules.

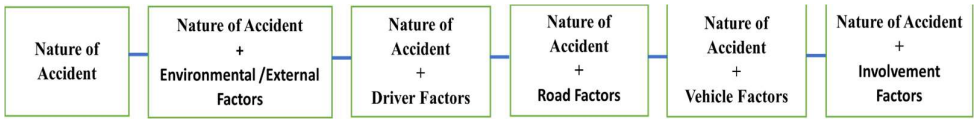

Fig 7 Main Groups of Attributes to Obtain Association Rules

The current study has used the metrics of support, confidence and lift to shortlist the most important rules. Keeping in view the volume of datasets, the threshold values of confidence and lift are taken as 0.5 and 1.0 respectively. The current study used a 2 fold cross validation method of n_splits=2, shuffle=True, random_state=SEED.

5 Results and Discussion

Asm we have obtained considerable knowledge about the data and can easily build models. To find an appropriate ML model for our proposed system, this study conducts two experiments on datasets using the sklearn and keras libraries of python. **sklearn and keras** are Python libraries that are used to train ML and Deep learning models, on the other hand apyori is a python library that utilizes the apriori algorithm and finds the associations. In this section, we investigate the following research question:

Question 1: Which machine learning or deep learning model provides the best trade-off between accuracy and computational efficiency for injury classification

To address the above research question (i.e., Question 1) a combination of Deep Learning (DL) and Machine Learning (ML) models has been employed on KSI dataset.

To explore the current study research question, we check the performance of different deep Learning and Machine learning models that are Random Forest, SVM, XGBoost, BiLSTM, ATTR-LSTM, KNN-LSTM, and W-CNN-LSTM on KSI dataset using their default and hyperparameters. The current study Select these models because their performance is good on both numeric and categorical data. Random Forest & XGBoost are ensemble-based models and handle any mix data type, furthermore it can handle model overfitting problems [18] [27]. Support Vector Machine (SVM) performance is good on high -dimensional data, it used decision boundary concept during classification task [28], The current study used Deep Learning Models (BiLSTM, ATTR-LSTM, KNN-LSTM, W-CNN-LSTM) because it suitable for sequential nature of our dataset. The LSTMS model is famous for capturing dependencies and contextual relationships between data [29] (. W-CNN-LSTM is a new version of the LSTM model which increases the accuracy and improves feature extraction through convolutional features. The current study combines both traditional ML and Deep learning models to ensure the effective analysis of injury severity classification.

To highlight the strength and weaknesses of every classification model the confusion matrices have been used as shown in **Fig. 8-14**. The KSI dataset is used for classifying the injury severity.

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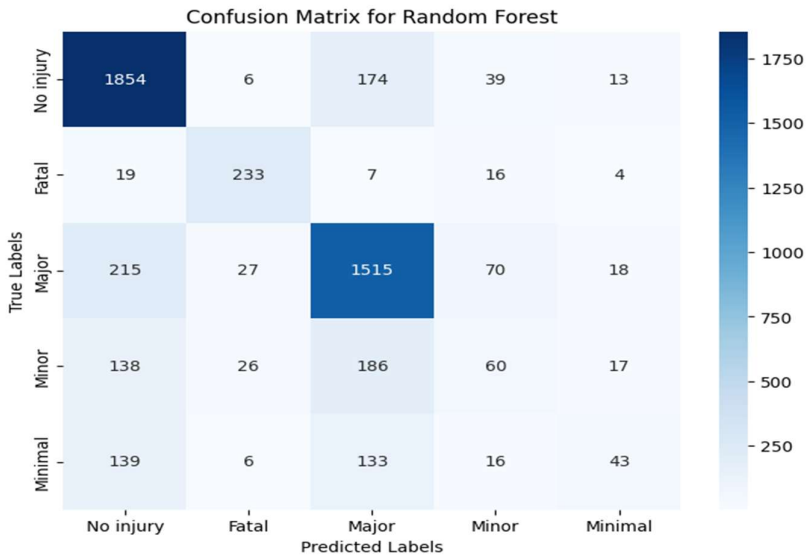

Fig 8 Confusion Matrix of Random Forest Model on KSI Dataset

Classification Report

	precision	recall	f1-score	support
No injury	0.78	0.89	0.83	2086
Fatal	0.78	0.84	0.81	279
Major	0.75	0.82	0.78	1845
Minor	0.30	0.14	0.19	427
Minimal	0.45	0.13	0.20	337
accuracy			0.74	4974
macro avg	0.61	0.56	0.56	4974
weighted avg	0.71	0.74	0.72	4974

Fig 9 Class wise Performance of Random Forest Model on KSI Dataset

First, we trained Random Forest (RF) on default parameters (n_estimators=100, random_state=42) on KSI dataset. The **Fig. 8,** show that true positive rate of Random Forest on no Injury=0.8889 (88.89%), Fatal=0.8351 (83.51%), Major=0.8211 (82.11%), Minor=0.4357 (43.57%), Minimal=0.1274 (12.74%). The result shows that the model performs well in identifying No Injury (88.89%) and Fatal cases (83.51%). The performance of RF models across different classes shown in **Fig. 9**.

Major injuries (82.11%) are also classified relatively well, but the result shows that the model struggles with Minor (43.57%) and Minimal injuries (12.74%), likely due to class imbalance or overlapping features.

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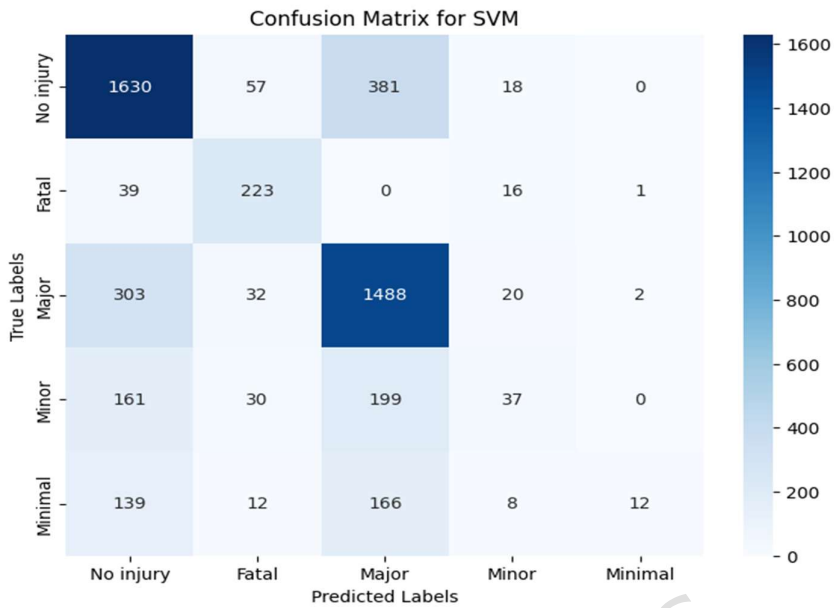

Fig 10 Confusion Matrix of SVM Developed Over KSI Dataset

Classification Report

	precision	recall	f1-score	support
No injury	0.72	0.78	0.75	2086
Fatal	0.63	0.80	0.70	279
Major	0.67	0.81	0.73	1845
Minor	0.37	0.09	0.14	427
Minimal	0.80	0.04	0.07	337
accuracy			0.68	4974
macro avg	0.64	0.50	0.48	4974
weighted avg	0.67	0.68	0.64	4974

Fig 11 Class wise Performance of SVM Model on KSI Dataset

In the second phase, we trained SVM on default parameters (kernel='rbf', probability=True) on KSI dataset. The **Fig. 10** show that True positive rate of SVM on no **Injury**=0.7814(78.14%), **Fatal=0.7993** (79.93%) ,Major=**0.8062** (80.62%),Minor=**0.4660** (46.60%),Minimal=**0.0352** (3.52%). The Result shows that the SVM model performs well in identifying No Injury, Fatal, and Major cases. Moreover, class wise performance of the SVM model on KSI dataset shown in **Fig. 11**.

The SVM model struggles with **Struggles with Minor (46.60%) and Minimal (3.52%) cases**, indicating possible class imbalance or feature overlap

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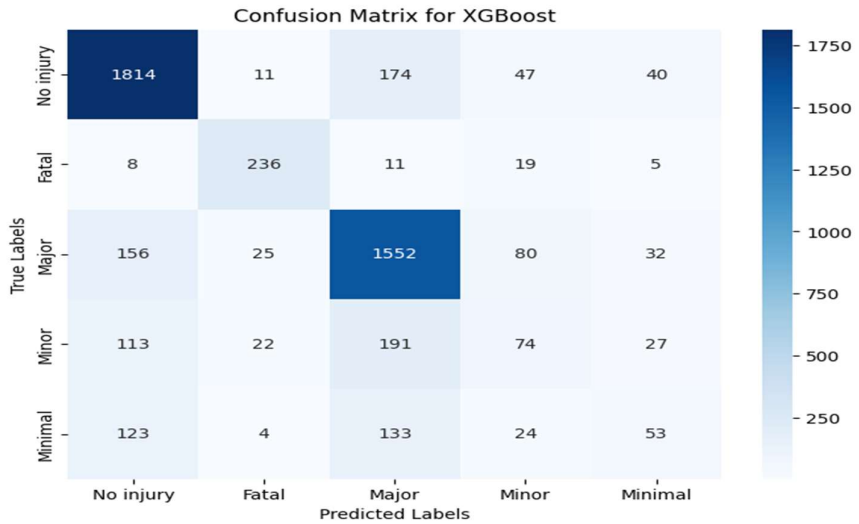

Fig 12 Confusion Matrix of XGBoost on KSI Dataset

Classification Report

	precision	recall	f1-score	support
No injury	0.82	0.87	0.84	2086
Fatal	0.79	0.85	0.82	279
Major	0.75	0.84	0.79	1845
Minor	0.30	0.17	0.22	427
Minimal	0.34	0.16	0.21	337
accuracy			0.75	4974
macro avg	0.60	0.58	0.58	4974
weighted avg	0.72	0.75	0.73	4974

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Fig 13 Class wise Performance of XGBoost Model on KSI Dataset

Third, we trained XGBoost on default parameters (use_label_encoder=False, eval_metric='mlogloss') on KSI dataset..True positive rate of XGBoost on **no Injury=0.8695** (86.95%), **Fatal= 0.8459** (84.59%), **Major=0.8411** (84.11%),Minor=**0.4471** (44.71%), Minimal=**0.1572** (15.72%). The **Fig. 12** shows that the XGBoost model performs well in identifying No Injury, Fatal, and Major. The class wise performance of XGBoost model presented in **Fig. 13**.

The XGBoost model struggles with **Minor (44.71%) and Minimal (15.72%) cases, similar to SVM, but slightly better than SVM for Minimal cases.**

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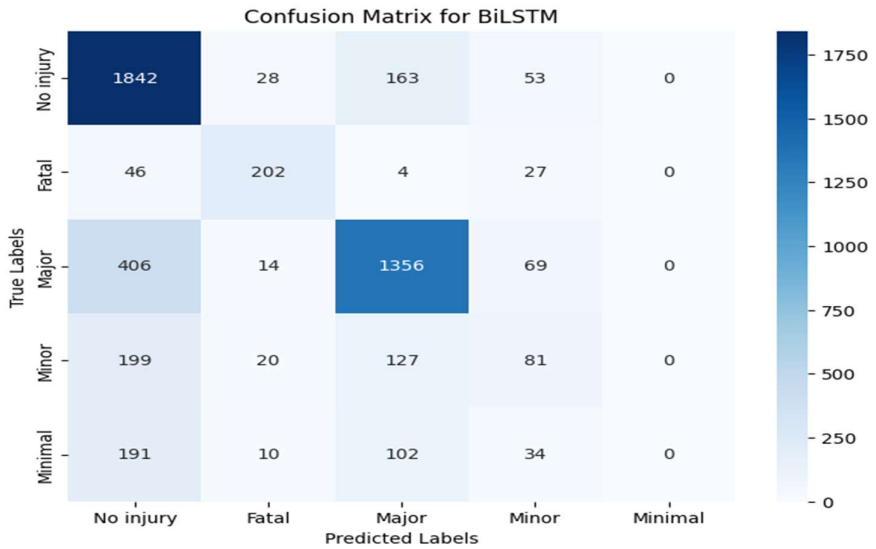

Fig 14 Confusion Matrix of BiLSTM Developed Over KSI Dataset

Classification Report

	precision	recall	f1-score	support
No injury	0.75	0.81	0.78	2086
Fatal	0.69	0.84	0.75	279
Major	0.69	0.84	0.76	1845
Minor	0.32	0.09	0.14	427
Minimal	0.00	0.00	0.00	337
accuracy			0.71	4974
macro avg	0.49	0.51	0.49	4974
weighted avg	0.63	0.71	0.66	4974

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Fig 15 Class wise Performance of BiLSTM Model on KSI Dataset

Fourth, we trained BiLSTM on parameters (**activation='relu','softmax', dropout (0.3**) on KSI dataset. True positive rate of BiLSTM on **no Injury=0.8830** (88.30%), **Fatal= 0.7239 (72.39%)**, **Major=0.7350** (73.50%), Minor=**0.2977** (29.77%), Minimal= **0.00%**. The **Fig.14** that BiLSTM model performs well in identifying "No Injury" (88.30%) but drops significantly for "Minimal" (0.00%) cases. Major and Fatal categories are classified moderately well (73.50% and 72.39%) but are lower compared to XGBoost. The BiLSTM model struggles **with Minor injuries (29.77%). Fig. 15** shows performance of BiLSTM across different classes.

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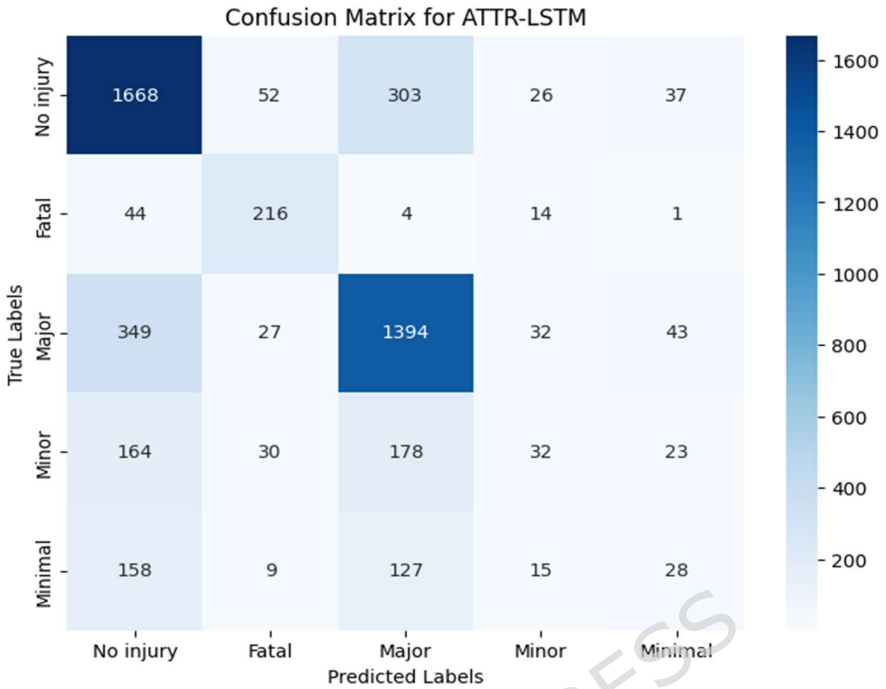

Fig 16 Confusion Matrix of ATTR-LSTM Developed Over KSI Dataset

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Classification Report

	precision	recall	f1-score	support
No injury	0.68	0.81	0.74	2086
Fatal	0.71	0.70	0.70	279
Major	0.70	0.74	0.72	1845
Minor	0.27	0.15	0.19	427
Minimal	0.23	0.02	0.04	337
accuracy			0.67	4974
macro avg	0.52	0.49	0.48	4974
weighted avg	0.63	0.67	0.64	4974

Fig 17 Class wis**e** Performance of ATTR-LSTM Model on KSI Dataset

Then we trained ATTR-LSTM on parameters (**activation='softmax', flatten layer, return_sequences=False**) on KSI dataset. The **Fig. 16** show that True positive rate of ATTR-LSTM on no **Injury=0.7997** (79.97%), **Fatal= 0.7741 (77.41%)**, **Major=0.7558** (75.58%), Minor=**0.4168** (41.68%), Minimal= **0.0835 (8.35%)**. The Result shows that the ATTR-LSTM performs better than BiLSTM in most categories."No Injury" classification is slightly lower than BiLSTM (79.97% vs. 88.30%)."Fatal" and "Major" TPRs are higher than BiLSTM (77.41% vs. 72.39% and 75.58% vs. 73.50%)."Minimal injury detection is slightly better than BiLSTM (8.35% vs. 0.00%) but still very low. "Minor" class also improved (41.68% vs. 29.77%). **Fig. 17** shows performance of ATTR-LSTM across different classes on KSI dataset.

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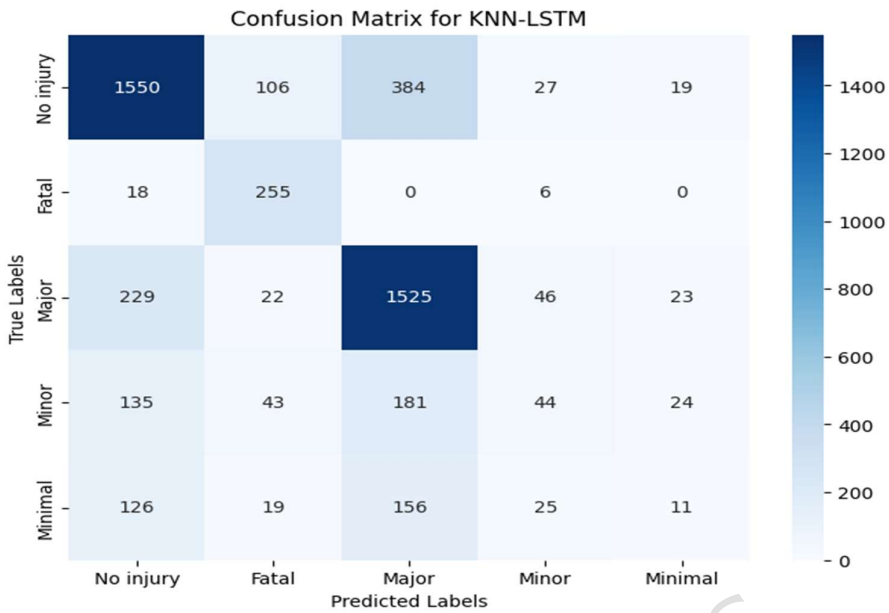

Fig 18 Confusion Matrix of KNN-LSTM Developed Over KSI Dataset

Classification Report

	precision	recall	f1-score	support
No injury	0.77	0.75	0.76	2086
Fatal	0.58	0.86	0.69	279
Major	0.66	0.84	0.74	1845
Minor	0.35	0.12	0.18	427
Minimal	0.29	0.01	0.01	337
accuracy			0.69	4974
macro avg	0.53	0.52	0.48	4974
weighted avg	0.65	0.69	0.65	4974

Fig 19 Class wise Performance of KNN-LSTM Model on KSI Dataset

Then we trained **KNN-LSTM** on parameters (**activation='softmax', 'relu', flatten layer,return_sequences=False**) on KSI dataset.. True positive rate of **KNN-LSTM** on no **Injury=0.7431** (74.31%), **Fatal= 0.9140 (91.40%))**, **Major=0.8263** (82.63%), Minor=**0.4242** (42.42%), Minimal= **0.0325 (3.25%)**.

The Result in **Fig. 18** shows that KNN-LSTM achieves highest accuracy for Fatal cases (91.40%) and Major cases (82.63%) and No Injury class performance is slightly lower than ATTR-LSTM (74.31% vs. 79.97%). Minimal class has a very low recall (3.25%), similar to other models. Minor class TPR (42.42%) is slightly better than ATTR-LSTM (41.68%). Moreover, the class-wise performance of KNN-LSTM shown in **Fig. 19**.

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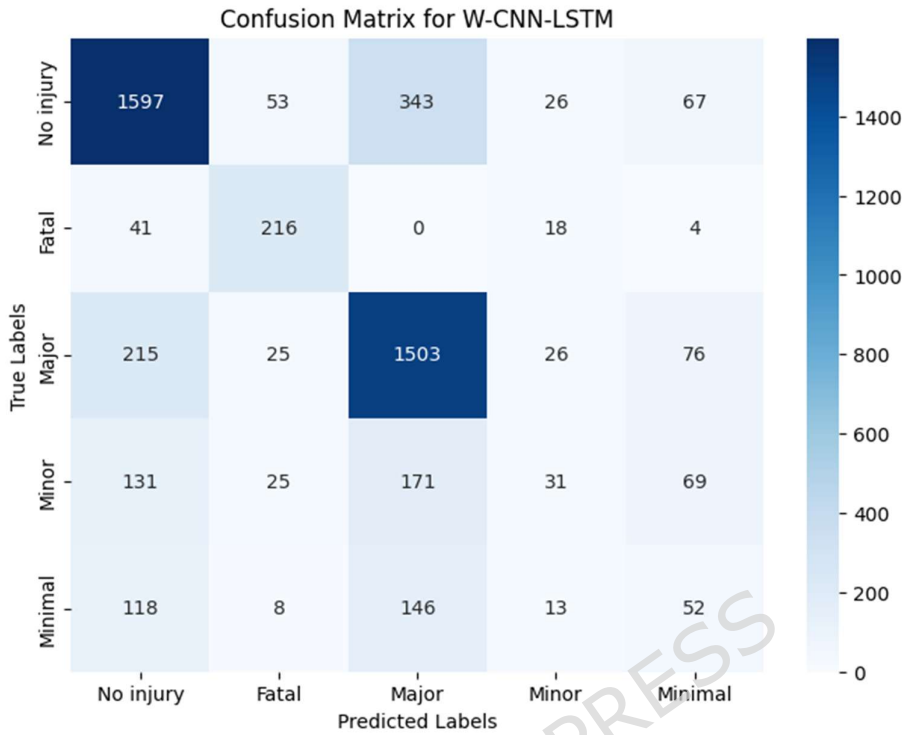

Fig 20 Confusion Matrix of W-CNN-LSTM Developed Over KSI **Dataset**

Classification Report

	precision	recall	f1-score	support
No injury	0.67	0.86	0.76	2086
Fatal	0.62	0.79	0.70	279
Major	0.74	0.73	0.74	1845
Minor	0.29	0.07	0.11	427
Minimal	0.24	0.02	0.04	337
accuracy			0.68	4974
macro avg	0.51	0.49	0.47	4974
weighted avg	0.63	0.68	0.64	4974

Fig 21 Class wis**e** Performance of KNN-LSTM Model on KSI Dataset

Finally, we trained W-CNN-LSTM on parameters (**n_estimators=100, random_state=42**) on KSI dataset. True positive rate of W-CNN-LSTM on **no Injury=0.7655** (76.55%), **Fatal= 0.7741 (77.41%)**, **Major=0.8147** (81.47%), Minor=**0.4009** (40.09%), Minimal= **0.1545 (15.45%)**.

The Result in **Fig. 20** shows that W-CNN-LSTM achieves the highest accuracy No Injury (76.55%) and Major (81.47%) on KSI dataset. The fatal class has a lower TPR (77.41%) compared to KNN-LSTM (91.40%) on KSI dataset. Minimal class has a better recall (15.45%) compared to KNN-LSTM (3.25%) but still needs improvement. Minor class performance is weaker (40.09%) than KNN-LSTM (42.42%). **Fig. 21** Class wis**e** Performance of W-CNN-LSTM Model on KSI Dataset.

- The BiLSTM shows higher performance in identifying "No Injury" and Major Injuries however, BiLSTM leads to frequent misclassification in distinguishing Minimal and Minor injury categories.The ATTR-LSTM comparatively

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performs better than BiLSTM for the identity of Major injuries and leading higher misclassification in Minor and Minimal injuries. KNN-LSTM performs well for Major and Fatal categories of injuries while shows poor performance in classifying Minimal and Minor injuries. W-CNN-LSTM demonstrates balanced results regarding most of the injury categories with good accuracy. However, results in misclassification for Minor category of injuries. The Random Forest (RF) classification algorithm shows better performance for dominant classes and lower performance for categories with lower frequency i.e., class imbalance remains a challenge for RF models. The traditional models like SVM show poor performance as compared to Deep Learning (DL) models. This is because these models lack the capability to effectively handle sequential dependency. In summary, as compared to the traditional models, the DL models especially the W-CNN-LSTM shows better performance due to their hierarchical extraction of features. However, there is still a need to enhance the capability of DL models for refined feature representation and to address class imbalance issues.

Table 6- Models Accuracy Comparison on KSI dataset

Model	**Accuracy**	**Precision**	**Recall**	**f1-score**
Random Forest	0.7449	0.71	0.74	0.72
SVM	0.6815	0.67	0.68	0.64
XGBoost	0.7497	0.72	0.75	0.73
BiLSTM	0.6998	0.64	0.70	0.66
ATTR-LSTM	0.6711	0.63	0.67	0.64
KNN-LSTM	0.6805	0.64	0.68	0.65
W-CNN-LSTM	0.6834	0.65	0.68	0.66

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The evaluation of various machine learning (ML) and deep learning models on injury severity classification reveals distinct performance trends (as shown in Table 6). Among traditional ML models, **XGBoost (accuracy: 0.7497, F1-score: 0.73) and Random Forest (accuracy: 0.7449, F1-score: 0.72)** emerged as the top-performing models on KSI dataset., demonstrating strong predictive capabilities. These models effectively handle categorical and numerical data, leveraging ensemble learning techniques to enhance classification performance [27] [18].

On the other hand, deep learning models, including **BiLSTM, ATTR-LSTM, KNN-LSTM, and W-CNN-LSTM**, exhibited moderate accuracy ranging from **0.6711 to 0.6998 on** KSI dataset. While these models have the advantage of capturing sequential patterns in data, their performance was slightly lower than XGBoost and Random Forest, possibly due to the dataset's complexity and size. Among deep learning models, **BiLSTM and W-CNN-LSTM** performed slightly better, suggesting that CNN-based feature extraction and bidirectional learning improve model effectiveness. Support Vector Machines (SVM) demonstrated the lowest accuracy (**0.6815**) on KSI dataset. among traditional ML approaches, indicating that linear decision boundaries may not be optimal for this dataset. However, SVM still showed competitive precision and recall values, implying its potential effectiveness with additional feature engineering.

Overall, **XGBoost and Random Forest emerged as the most reliable models**, offering a balance between precision, recall, and F1-score. While deep learning models show promise, further optimization, larger datasets, or feature extraction techniques could enhance their predictive power. Deep learning models like KNN-LSTM, W-CNN-LSTM, BiLSTM, and ATTR-LSTM has shown competitive AUC values (ranging from .93 to .91) demonstrating their potential for complex pattern capturing. However, their performance can be improved further by adding more data.

The comparison results shown in Fig. 15 reveals that the XGBoost outperform in terms of ROC curve against their counterparts. The XGBoost, RF, and BiLSTM achieved .95, .94, and .93 on KSI dataset for AUC respectively. These results show that XGBoost and RF can be considered as the most favorite classifiers for the mentioned dataset.

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Due to the limited temporal coverage, the KSI dataset leads to major challenges in applications of DL models, specifically in terms of generalization and model training. Because DL models need large data and memory to effectively capture the underlying patterns. Research studies show that due to inadequate coverage of phenomena, small datasets result in model overfitting which degrades the performance of DL models as it memorizes the training data instead of generalizing it [33]. Moreover, due to the limited scope of temporal data, the adaptability of DL models to the changes over time which is highly important for most of the real-world applications [34]. Therefore, development of robust and reliable models on KSI dataset could become a challenge of DL models.

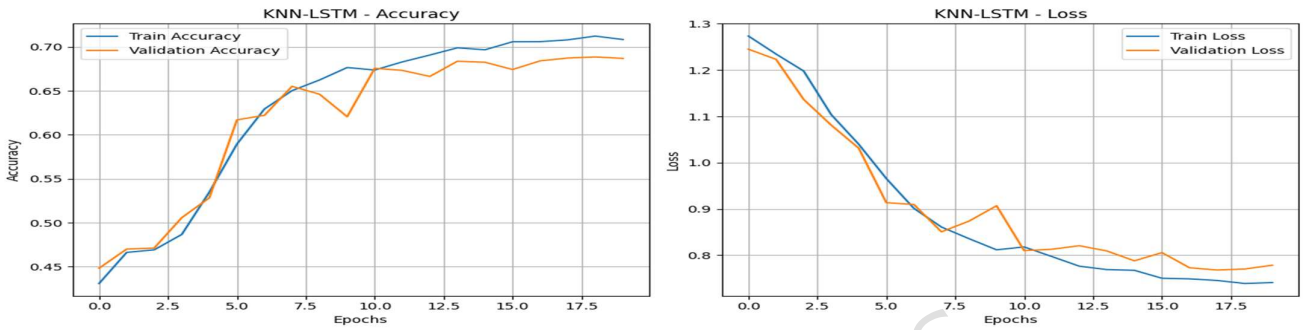

Fig 22 Learning curves of KNN-LSTM

Fig. 22 left graph shows the accuracy curve for both training and validation accuracy across 20 epochs. The accuracy indicates a steady improvement during the initial epochs, with both training and validation accuracy increasing from approximately 0.45 to above 0.65 by the tenth epoch. The training and validation accuracy rise steadily together, which indicates that the model has learned useful patterns without overfitting in the early stage. Furthermore, validation accuracy quickly jumps closer to the training accuracy by epoch 5. The right graph shows a loss curve which demonstrates a similar trend, with training and validation loss decreasing in the early epochs consistently, followed by slow convergence after epoch 10. While training loss continues to decline to approximately 0.75, validation loss stabilizes around 0.78–0.80, suggesting mild overfitting. Overall, the results show that the KNN-LSTM model converges effectively, achieves stable performance, and demonstrates reasonable generalization ability. Both curves gradually converge, with training accuracy stabilizing around **0.71–0.72** and validation accuracy around **0.68–0.69**.

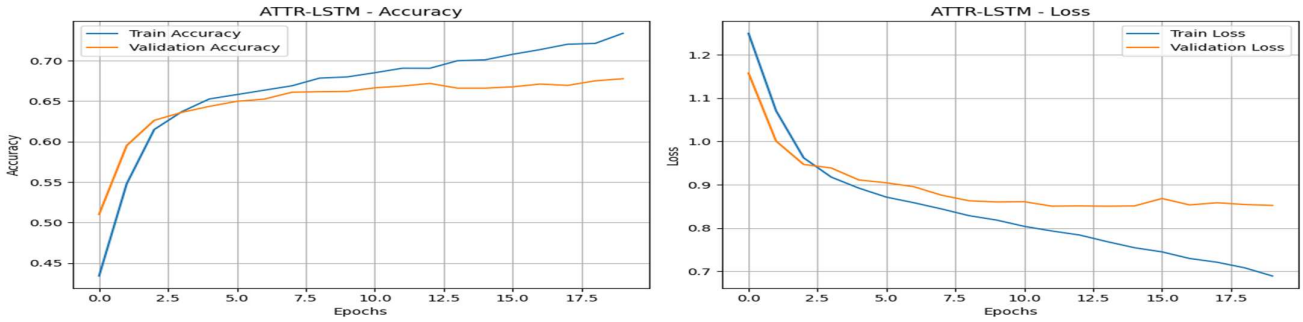

Fig 23 Learning curves of ATT-LSTM

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Fig. 23, ATTR-LSTM achieved higher validation accuracy, but its performance plateaued around epoch 10 (~0.66–0.67), whereas training accuracy continued to improve to **~0.72**. Similarly, validation loss in ATTR-LSTM is static and even fluctuated upward.

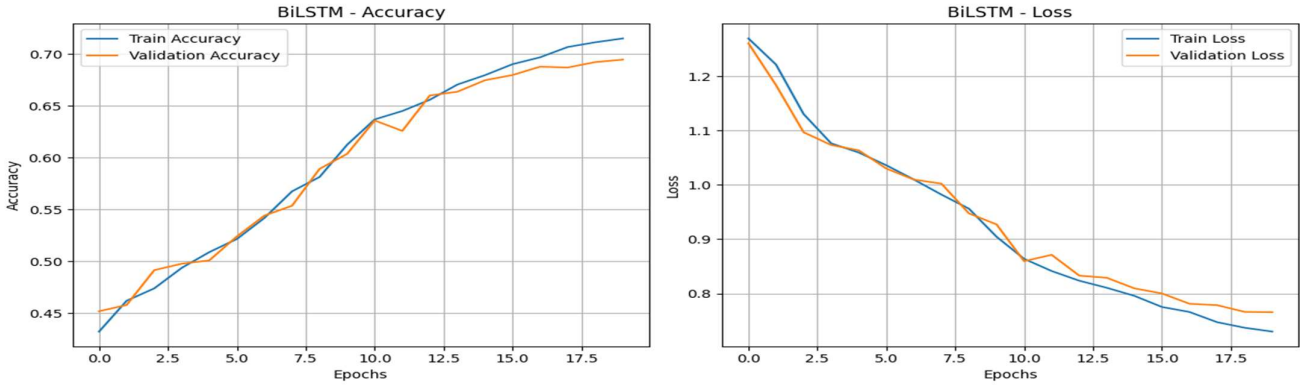

Fig 24 Learning curves of BiLSTM

Fig. 24, BiLSTM exhibited consistent learning behavior with strong generalization performance across training and validation sets. The accuracy curve with training accuracy reaching approximately **0.72** and validation accuracy closely following at around **0.70**, indicating minimal divergence. Similarly, the loss curve demonstrated smooth convergence, with both training and validation loss.

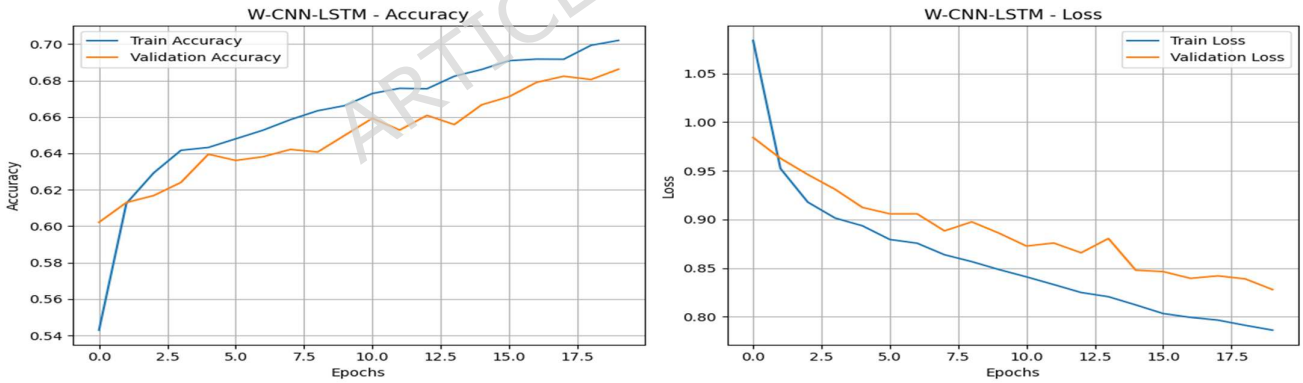

Fig 25 Learning curves of W-CNN-LSTM

Fig. 25, W-CNN-LSTM has trained for 20 epochs where the training accuracy increases smoothly from ~0.54 → ~0.70; validation accuracy increases from ~0.60 → ~0.69. Similarly, training loss declines steadily from ~1.08 → ~0.79; validation loss declines from ~0.98 → ~0.83 indicating the model is learning from the training data.

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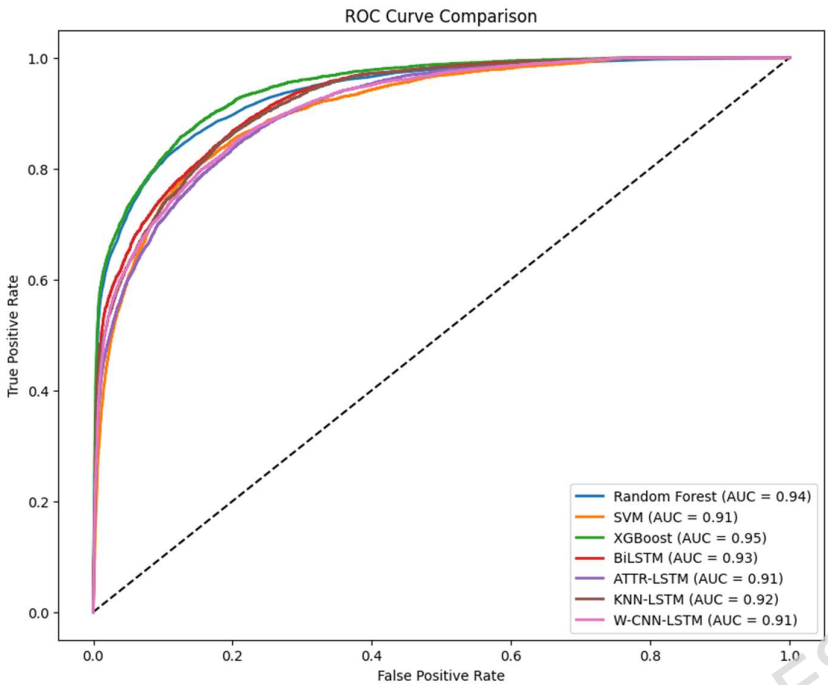

Fig 26 ROC Curve of Top Performing ML Models Trained over KSI Dataset

The W-CNN-LSTM, ATTR-LSTM, and SVM have shown comparable performance on KSI dataset by obtaining an AUC of 0.91 (as shown in **Fig. 26**). However, the ensembled based models like RF and XGBoost showed slightly better performance. These results reveal that ensemble Learning models are more reliable for handling structured datasets and with categorical attributes.

To explore the answer of research question 1, In this study, XGBoost and Random Forest achieved high accuracy in classified injury types. XGBoost demonstrated strong performance with accuracy (0.75) and an AUC of 0.75 values because these models are well suited for structure and categorical data. Similarly, Random Forest received accuracy of 0.74 and higher AUC of 0.94 which is near to previous models. Additionally, these models have lower computation times as compared to deep learning models.
The BiLSTM achieved good performance (accuracy: 0.70, AUC: 0.93) compared to other deep learning models on **the** KSI dataset. This finding indicates that BiLSTM effectively captures complex temporal patterns. On the other side, **ATTR-LSTM, KNN-LSTM, and W-CNN-LSTM** show low performance in predicting injury type with accuracy ranging from **0.67 to 0.68** and AUC values **0.91 to 0.92**. These results recommend that ensemble- based model XGBoost and Random Forest are promising choices to address this problem.
After training and testing Machine learning and deep learning models with default configuration, hyperparameter tuning is performed to enhance the performance of the trained models. For this purpose, a Grid search method has been used which evaluates multiple combinations of hyperparameters in a systematic way. The aim of using the Grid search method is to identify the best combination of hyperparameters that provide higher performance on the validation of datasets. To ensure robustness and reduce overfitting, a 2-fold cross validation with seed =42 has been used. The experimental setup for hyperparameter tuning shown in Table 5.
The performance of trained models with default parameters has been compared with the hyperparameter tuned performance of these models (as shown in **Table 7**).

Table 7- Comparison of Models Performance on Default and Tuned Parameters

Model	**Accuracy** (Default)	**Accuracy** (Tuned)	**F1-Score** (Default)	**F1-Macro** (Tuned)	**F1-Micro** (Tuned)	**Parameters for tuned model**

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Random Forest	0.7449	0.9963	0.72	0.9942	0.9963	max_depth: none, n_estimators: 50
SVM	0.6815	0.3568	0.64	0.1572	0.3568	C: 1, kernel: rbf
XGBoost	0.7497	0.7724	0.73	0.574	0.7724	max_depth: 3, n_estimators: 50
BiLSTM	0.6998	0.384	0.66	0.358	0.384	Unites: 32, dropout: 0.3, epoch:8
ATTR-LSTM	0.6711	0.53	0.64	0.46	0.53	Unites: 32, dropout: 0.3, epoch:8
KNN-LSTM	0.6805	0.38	0.65	0.35	.38	Unites: 32, dropout: 0.3, epoch:8
W-CNN-LSTM	0.6834	0.53	0.66	0.47	0.53	Unites: 32, dropout: 0.3, epoch:8

The hyperparameter tuning results show that the performance of the Random Forest (RF) model substantially improved for accuracy, and F1-score as compared to SVM and XGBoost. The RF model has obtained 0.9963, 0.9942, and 0.9963 accuracy, F1-macro, and F1-micro respectively. This highlights the capability of RF to effectively handle class imbalance and to capture complex features interactions. The XGBoost demonstrates moderate improvement for hyperparameter tuning as compared to the default parameters in terms of accuracy and F-1 micro score. However, obtained a lower F-1 macro score which advocates that the XGBoost model faces difficulty in balancing minority class predictions. Similarly, SVM has shown drastic decline in the performance for all the three parameters which reveals their limitations in handling feature distribution of datasets and sensitivity to hyperparameter tuning.

On the other hand, the hyperparameter tuning results show that the performance of deep learning models like BiLSTM, ATTR-LSTM, KNN-LSTM, and W-CNN-LSTM has decreased after tuning. This decrease in the performance can be associated with limited training epochs, shallow hyperparameter configurations, and small dataset size that restrained their capability to learn complex representations.

The findings of hyperparameter tuning suggests the effectiveness and reliability of the machine learning models like RF and XGBoost for structured accident data. Moreover, to achieve competitive performance for deep learning models, there is need for extended training, larger datasets, and more advanced tuning strategies.

Question 2: What are the underlying hidden factors that contribute to fatal or major injuries?

To answer the second research questions and identify patterns of factors, a second experiment was conducted using association rule mining (ARM). This analysis was performed on both the KSI and RATA dataset. This experiment utilized Python and apriori algorithms and 128 rules were extracted but only 6 rules were important in KSI dataset and 4 rules were important in RTA dataset (as shown in **Fig. 27**) which is described below:

Rule 1: indicates a strong association between accidents involving pedestrians and the absence of red-light violations on weekends. The higher lift value shows a stronger association between antecedent and

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consequence, and the confidence value shows that there are 97% chances of pedestrian collisions on weekends with no red-light violation. The support, confidence and lift on the subgroup nature of accident are mentioned in **Table 8**:

Table 8: Top Associations for Nature of Accident over KSI

Rule	Antecedent	Consequent	Support	Confidence	Lift
1	Pedestrian Collisions	No Red-light Violation, Weekend	0.05	0.97	3.85
2	Speeding	Aggressive Driving	0.12	0.85	2.42
3	Rear End, No Red-light Violation	Non-Fatal, Aggressive Driving	0.06	0.70	2.26
4	Red-light Violation	No Over speeding, Non-Fatal Injury, Aggressive Driving	0.062	0.73	2.13
5	No Over speeding, Pedestrian Collisions	None	0.19	0.95	1.12
6	SMV	No Red-light Violation, Non-Fatal	0.06	0.82	1.04

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Rule 2: suggests a strong association between speeding and aggressive driving. The support indicates that this pattern is present in 12% of the dataset. The confidence value shows that when there is speeding, there is an 85% likelihood of aggressive driving. The higher lift value suggests that the occurrence of aggressive driving is 2.42 times more likely when speeding is observed, compared to when these events are independent.

Rule 3: indicates that aggressive driving is the reason for 70% of rear-end collisions regardless of the red-light violations, and these collisions lead to non-fatal accidents. The higher lift value shows a stronger association between these attributes.

Rule 4: indicates red light violations are associated with the absence of over speeding, coupled with aggressive driving that leads to non-fatal injuries. The support shows that this pattern occurs in 6.2% of the dataset. The

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confidence value suggests that when there is a red-light violation, there is a 73.3% likelihood of observing no over speeding and aggressive driving.

Rule 5: shows a mild association between accidents involving pedestrians with the injury severity as none. The rule also states that about 95% of such collisions exist in the dataset with no over speeding from the vehicle involved.

Rule 6: shows that the Slow-Moving Vehicle (SMV) collisions lead to non-fatal injuries. The confidence value shows that about 82% of collisions are of this nature, furthermore, these collisions do not break the red-light violation.

The same association rule mining approach was applied to the RTA dataset during covid-19 period.

Rule 1: within the 'Nature of Accident' over RTA dataset shows an association between over speeding during a time window of 12-18 hours (12:00 PM to 6:00 PM) and minor, non-fatal accidents. The confidence value 0.57 suggests that 57% of accidents involving over speeding, in this time range, were classified as minor and non-fatal.

Rule 2: highlights a strong association between careless driving behavior and single fractures with non-fatal outcomes. The very high confidence of 0.99 indicates that in nearly all the accidents involving carelessness and result the consequence being a single fracture and non-fatal.

Rule 3: within 'Nature of Accident and External Factors' category over RTA dataset shows a strong association between accidents occurring during the night hours (18 to 24:00) or (6:00 PM to 12:00 AM) and the outcome being a single fracture. The confidence value of 0.1 shows that, in the dataset, all accidents recorded under these conditions were associated with single fractures.

Rule 4: shows that the injury severity of minor, non-fatal accidents are more common during rainy days between 12 PM and 6 PM. The confidence value of 0.78 shows that when accidents happen during the specified time range on rainy days, there is a 78% chance of minor and non-fatal injuries.

The rule 1 within 'Nature of Accident and Vehicle Factors' over RTA dataset shows strong association between over speeding on motorcycles during weekends and minor injury severity. The rule 2 shows association between over speeding in cars and rickshaws results in non-fatal and minor injuries. The confidence value of 0.64 and 0.71 of rule 2 shows that 64% and 71% chances of non-fatal and minor injuries respectively. The support, confidence and lift on the subgroup nature of accident are mentioned in **Table 9**:

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Table 9: Top Associations for Nature of Accident over RTA

Rule	Antecedent	Consequent	Support	Confidence	Lift
1	12 to 18, Over Speed	Minor, Non-Fatal	0.14	0.57	1.04
2	Carelessness	Single Fracture, Non-Fatal	0.05	0.99	1.01
3	18 to 24, Night	Single Fracture	0.04	1	2.88
4	12 to 18, Rainy, Day	Minor, Non-Fatal	0.05	0.78	1.55

The RTA dataset corresponds to the COVID-19 pandemic period and reflects road usage during this time. The results show that the overall traffic volume during weekends is low however, vehicle speed is high. Moreover, motorcycle usage and increased speeds due to limited public transport on weekends results in a higher number of motorcycle accidents. These findings suggest the implementation of stricter government policy on speed regulation during low-traffic periods and highlights the importance of targeted interventions for high-risk groups. Furthermore, such policies should remain flexible to allow timely adjustments during crisis situations.

The current study examines traffic characteristics and accident trends in Pakistan and Canada, highlighting the needs for country specific interventions that accounts for differences in traffic behaviors and road infrastructure. In Canada, well developed infrastructure contributes to lower reliance on motorcycles while Pakistan faces more challenges due to high motorcycle usage. To address these challenges, both countries need to implement effective road safety policies.

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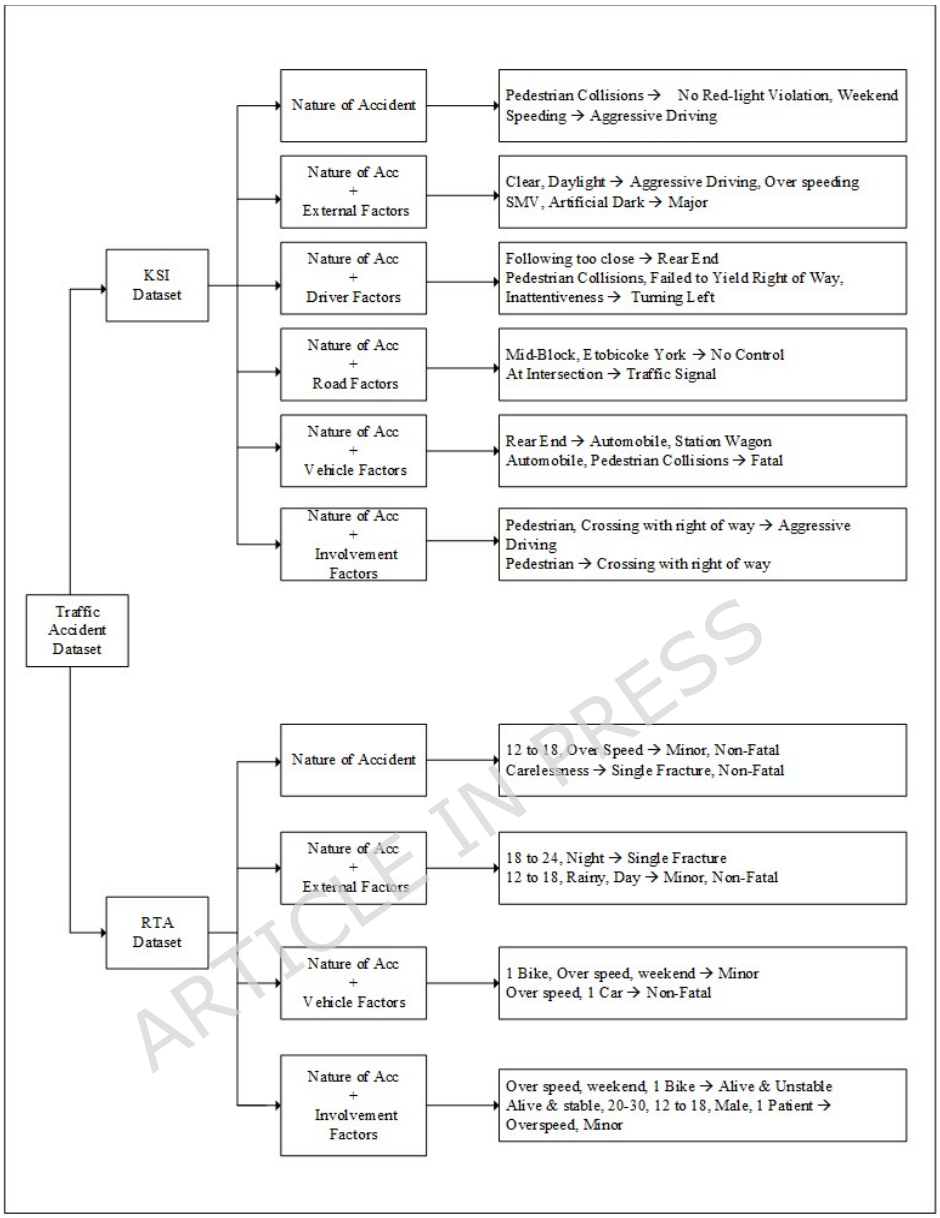

Fig 27 Top Association Rules of KSI and RTA Datasets

6. Conclusion and Future Work

Currently, prediction of road traffic accidents injuries and their severity has become an important challenge for traffic control departments around the globe. Researchers have used many AI and machine learning techniques to solve various challenges when attempting to construct a warning system that can predict the traffic alert.

The results show that **XGBoost Classifiers** are the top three best injury severity predictors for the KSI dataset. in terms of accuracy, recall and AUC. However, if the metric of accuracy and recall is considered then the XGBoost model has achieved the highest recall of 75% and accuracy 74%. But after apply hyperparametering tuning, the performance of the RF model (Accuracy =0.99, F1-score=0.99) is increased as compared to other models.

Moreover, the researcher has conducted association rule mining techniques and identified various factors contributing to severe collisions in a developing country like Pakistan, as opposed to a developed nation like Canada. The results

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show that over speeding, carelessness and driver's inattentiveness can cause fatal or major injuries. However, some factors are more common in developing countries like Pakistan as compared to developed countries like Canada.
The results show that in Pakistan, the majority of the severe accidents involve young drivers below the age of 20 however, in Canada, the drivers involved in collisions with minimum age is between 20 and 24 years.

The current study also reveals that in Pakistan, fatal and major collisions of head injuries resulting from motorcycle accidents. On the other side, in Canada, major accidents often entail collisions between pedestrians or cyclists and automobiles.

Timing and rainy weather are important factors of severe collisions in Pakistan and Canada. In Pakistan, nighttime significantly contributes to higher probability of severe collisions compared to daytime incidents. In Canada, daytime emerges as a more critical period for major collisions. Similarly, in Pakistan, rainy weather is associated with a higher incidence of major accidents, whereas in Canada, fewer accidents occur during rainy conditions. This is due to the fact that drivers in Canada exhibit greater caution and restraint during rainfall, leading to a reduction in aggressive driving behaviors compared to their counterparts in Pakistan.

Based on the association rule mining results from the KSI (Canada) and RTA (Pakistan) datasets, several key recommendations can be made to improve road safety in both countries. In Canada, pedestrian safety can be enhanced by implementing stricter measures against aggressive driving and red-light violations, especially on weekends. Additionally, installing traffic control measures on Mid-Block roads in high-risk areas like District Etobicoke York could significantly reduce accidents. In Pakistan, over-speeding, particularly on motorcycles during weekends, is a major cause of minor but frequent accidents. Enforcing speed limits and increasing awareness about safe driving behaviors during peak accident hours (12 PM – 6 PM) can help mitigate risks. Moreover, improved road lighting and traffic signals at high-risk intersections could reduce the severity of nighttime accidents. Both countries should focus on targeted interventions, such as stricter enforcement of traffic laws, public awareness campaigns, and infrastructure improvements, to address their unique accident patterns effectively.

Limitations of the current Study:

This study successfully investigates road traffic accidents using Machine learning and association rule mining. However, the study has some limitations. First, the current study datasets have imbalance problems which may affect the model's generalization. Secondly, the current study focuses on two countries' datasets, which may not capture a global trend. Third, cultural, infrastructural, policy, and dataset differences were not explicitly modeled, yet these factors significantly influence traffic behaviors and may limit the generalizability of the study. Finally, due to the limitations of available computation resources, performance of deep learning models is not good which may be further fine-tuned with high computation resources in future.

This study successfully investigates road traffic accidents using machine learning and association rule mining. Nevertheless, several limitations should be acknowledged. First, the datasets used in this study suffer from class imbalance, which may affect the generalizability of the models. Second, the analysis is limited to datasets from two countries, which may not reflect broader global trends. Third, cultural, infrastructural, and policy differences were not explicitly modeled, yet these factors can significantly influence traffic behavior and may limit the real-world applicability of the findings. Finally, ensemble machine learning methods were not explored in this study, which could potentially enhance predictive performance in future research

Future Work:

In this research, the study has evaluated the performance of several machine learning classification algorithms on the KSI dataset. In the future, advanced machine learning or deep learning methods can be tried on KSI and RTA datasets with high computational resources for more robust hyperparameter tuning methods. There are different sampling techniques available, current study has implemented SMOTE, and in future we can implement some other sampling techniques like cluster-based oversampling, hybrid methods. We plan to explore cost-sensitive learning approaches and ensemble-based methods (e.g., Balanced Random Forest, class weighting strategies) to further improve the performance of minority classes. Furthermore, the study has used two datasets, and in future, study can implement different datasets from different countries to generalize the findings.

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Data Availability Statement: The current study data are publicly available online at (https://data.torontopolice.on.ca/pages/total-ksi) and (https://doi.org/10.7910/DVN/L51CR9) for research purposes. No participants' personal information (e.g., name or address) was included in this study.

Conflicts of Interest:

The authors have no conflicts of interest.

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Not Applicable.

Author contributions

Mushtaq Hussain and Muhammad Shujaat Abid designed the study. Muhammad Shujaat Abid, Mushtaq Hussain, and Said Nabi, performed conceptualization and Methodology. The software and analysis of data have been performed by Mushtaq Hussain, Yeong Hyeon Gu, and Dildar Hussain. Furthermore, Tayyaba Sehar, Mushtaq Hussain, Said Nabi wrote the original draft and visualized the figures. The final manuscript has been reviewed and edited by Dildar Hussain, Yeong Hyeon Gu, and Mushtaq Hussain. All authors have read and agreed to the published version of the manuscript.

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Declarations

Ethics approval and consent to participate

This study utilized data from the UNBC McMaster Shoulder pain Expression Archive Database, which is publicly available for research purposes. Ethical approval for the collection and use of this data was granted by the University of Northern British Columbia. Informed consent was obtained from participants, who were made aware of the study's objectives and their rights. Researchers can access the dataset by signing an agreement form and will receive instructions within five business days.

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